

12.120

EE25BTECH11002 - Achat Parth Kalpesh

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Question

Let $\mathbf{A}$ be a 4×4 matrix with eigen values $-5, -2, 1, 4$. Which of the following is the eigenvalue of

$$\begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix}, \quad (1)$$

where $\mathbf{I}$ is the 4×4 identity matrix ?

① -5

② -7

③ 2

④ 1

Solution

Let $\mathbf{v}$ be any eigenvector of $\mathbf{A}$ with eigenvalue λ , i.e.

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \quad (2)$$

and

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix} \quad (3)$$

Let

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \quad (4)$$

be an eigenvector of $\mathbf{M}$ corresponding to eigenvalue μ where $\mathbf{x}_1$ and $\mathbf{x}_2$ are 4×1 column vectors, i.e.

$$\mathbf{M}\mathbf{x} = \mu\mathbf{x} \quad (5)$$

Solution

Substituting,

$$\begin{pmatrix} \mathbf{A} & \mathbf{I} \\ \mathbf{I} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mu \mathbf{x}_1 \\ \mu \mathbf{x}_2 \end{pmatrix} \quad (6)$$

This gives two matrix equations:

$$\mathbf{A}\mathbf{x}_1 + \mathbf{x}_2 = \mu \mathbf{x}_1 \quad (7)$$

$$\mathbf{x}_1 + \mathbf{A}\mathbf{x}_2 = \mu \mathbf{x}_2 \quad (8)$$

From (7),

$$(\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1 + \mathbf{x}_2 = 0 \quad (9)$$

$$\mathbf{x}_2 = -(\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1. \quad (10)$$

Solution

Substitute this in (8):

$$\mathbf{x}_1 + \mathbf{A} [- (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1] = \mu [- (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1] \quad (11)$$

$$\mathbf{x}_1 - \mathbf{A} (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1 = -\mu (\mathbf{A} - \mu \mathbf{I}) \mathbf{x}_1 \quad (12)$$

$$\left[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2 \right] \mathbf{x}_1 = 0 \quad (13)$$

For a non-trivial solution,

$$\det \left[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2 \right] = 0 \quad (14)$$

As

$$\mathbf{A} \mathbf{v} = \lambda \mathbf{v} \quad (15)$$

$$\left[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2 \right] \mathbf{v} = \left[\mathbf{v} - (\mathbf{A} - \mu \mathbf{I})^2 \mathbf{v} \right] \quad (16)$$

Let

$$(\mathbf{A} - \mu \mathbf{I})^2 \mathbf{v} = (\mathbf{A} - \mu \mathbf{I})(\mathbf{A} - \mu \mathbf{I}) \mathbf{v} \quad (17)$$

$$(\mathbf{A} - \mu \mathbf{I}) \mathbf{v} = \mathbf{A} \mathbf{v} - \mu \mathbf{v} \quad (18)$$

$$= (\lambda - \mu) \mathbf{v} \quad (19)$$

$$(\mathbf{A} - \mu \mathbf{I})(\lambda - \mu) \mathbf{v} = (\lambda - \mu)^2 \mathbf{v} \quad (20)$$

$$[\mathbf{I} - (\mathbf{A} - \mu \mathbf{I})^2] \mathbf{v} = [1 - (\lambda - \mu)^2] \mathbf{v} \quad (21)$$

Solution

For a non-zero $\mathbf{v}$, this holds only if

$$1 - (\lambda - \mu)^2 = 0 \quad (22)$$

$$\mu = \lambda \pm 1 \quad (23)$$

Given eigenvalues of $\mathbf{A}$ are

$$\lambda \in \{-5, -2, 1, 4\} \quad (24)$$

Therefore, the eigenvalues of $\mathbf{M}$ are

$$\lambda + 1 \in \{-4, -1, 2, 5\} \quad (25)$$

$$\lambda - 1 \in \{-6, -3, 0, 3\} \quad (26)$$

Eigenvalues of $\mathbf{M}$ are $-6, -4, -3, -1, 0, 2, 3, 5$